

EECS 442 Homework 3
Programming assignment

Due: Thur 10.22.2009 at 3pm

This programming assignment is concerned with the idea of using RANSAC to reject outliers.

Write a function *Homography* that takes a set of at least 4 image point correspondences between two images (manually selected) and returns an estimate of the homographic transformation between the images. Use least square (SVD decomposition) to compute the homography. The images can be downloaded from here:

<http://www.eecs.umich.edu/~silvio/teaching/Homeworks/HW3/I1.png>
<http://www.eecs.umich.edu/~silvio/teaching/Homeworks/HW3/I3.png>

Then, write a function *Reject_Outliers* that takes a set of at least 50 *correct* image point correspondences between two images (inliers) and 10 *wrong* image point correspondences between two images (outliers), both manually selected. This gives rise to an inlier/outlier ratio $w=.83$. Use RANSAC to automatically detect the outliers. You may refer to Hartley and Zisserman for implementation details. Perform as many iterations (k) as needed in order to guarantee that algorithm selects a sample set with inliers only with probability $p=.95$. You may use the formula below:

$$k = \frac{\log(1 - p)}{\log(1 - w^n)}$$

What to return:

- Your code with comments
- Two original images on top of which you plot the correct 50 point correspondences using a green-cross mark.
- The same two images where you plot the 10 *wrong* image point correspondences between two images using a red-circle mark.
- A text file (generated by your code) where you report the last N iterations of RANSAC. At each iteration you need to print out a line reporting:

Iteration #
Number of outliers
Fitting error

- Report the line when RANSAC finds an outlier set of minimal cardinality (or, inlier set of max cardinality). This is the solution reached by the algorithm.
- Discuss your results